

Tweet Sentiment Analysis

Classifying Apple Product Tweets as Positive, Negative, or Neutral



The Business Problem



Stakeholder: Apple's Marketing & Product Design Team



Millions of tweets posted during product launches — impossible to read manually



Negative sentiment signals product issues that need fast attention



Teams need real-time brand health data to make informed decisions



Solution

An NLP classifier that automatically labels tweets as Positive, Negative, or Neutral — giving the team instant, scalable insight without manual review.

The Data



CrowdFlower Apple Twitter Sentiment Dataset

3,886

Total tweets

2014

Year collected

Kaggle

Source

Human-rated

Annotation method



Key challenge: only 11% positive tweets — class imbalance affects training

How We Prepared the Data

- 1 Map Labels** Converted numeric codes: 1 → Negative, 3 → Neutral, 5 → Positive
- 2 Clean Text** Removed URLs, @mentions, hashtags, punctuation, and numbers
- 3 Remove Stopwords** Filtered common English words (the, and, is...) using NLTK
- 4 Lemmatize** Reduced words to base form — running → run — using NLTK
- 5 Vectorize** Converted text to numbers: CountVectorizer (baseline) then TF-IDF

Libraries used: **re** (text cleaning) **nltk** (stopwords, lemmatization) **scikit-learn** (CountVectorizer, TfidfVectorizer, pipelines)

Modeling Approach

Four models, built iteratively from simple to complex:

01

Naive Bayes

Baseline — CountVectorizer,
fast and simple

02

Logistic Regression

TF-IDF features, balanced
class weights

03

LinearSVC

Strong text classifier, TF-IDF
with bigrams

04

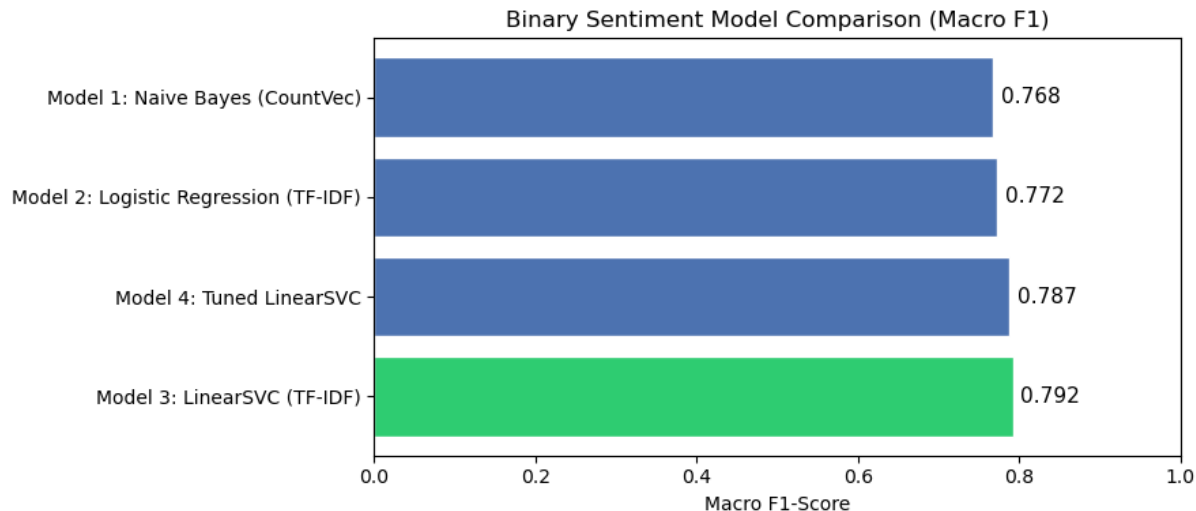
Tuned LinearSVC

GridSearchCV — optimal
regularisation + vocab size

Validation: Stratified 80/20 train-test split + 5-fold cross-validation for tuning — test set never seen during training

Results

Macro F1-Score by Model — Binary Task (Positive vs Negative):



0.79

Binary Macro F1
(Positive vs Negative)

0.67

Multiclass Macro F1
(3-class task)

Final model: Tuned LinearSVC

What the Model Learned

Top words driving each sentiment class (LinearSVC coefficients):

POSITIVE

love
great
best
amazing
awesome
nice
good
excellent
fantastic
happy

NEGATIVE

fail
sucks
broken
worst
hate
slow
annoying
bad
awful
crash

NEUTRAL

apple
iphone
aapl
stock
store
new
release
today
said
price

Limitations & Next Steps

Limitations

Only 11% of tweets are positive — minority class harder to classify

Tweets from 2014 — language and product names have evolved

Apple-only dataset — model needs retraining for other brands

No user or temporal context used in modeling

Next Steps



Collect more positive-class tweets to balance the dataset



Test transformer models (BERTweet) for improved short-text performance



Add sentiment:confidence as a feature weight during training



Deploy binary classifier as a real-time launch alert tool

Key Takeaways

Automated sentiment classification is achievable with high confidence

Macro F1 of 0.79 on binary task — ready for real-world testing

Model surfaces actionable vocabulary for product and campaign teams

A clear path to production exists — deploy as a launch alert tool

THANK YOU !!

QUESTIONS?

